# HealthTwin

- A synthetic-data digital twin prototype for T2D glucose-spike risk.
- Happiest Health Digital Twin Challenge submission
- Team and member details are pending confirmation.

# The problem

- A reactive alert arrives after glucose has crossed its threshold.
- Prototype question: can a two-hour spike risk be estimated and explained?
- Target event: glucose >180 mg/dL within two hours.

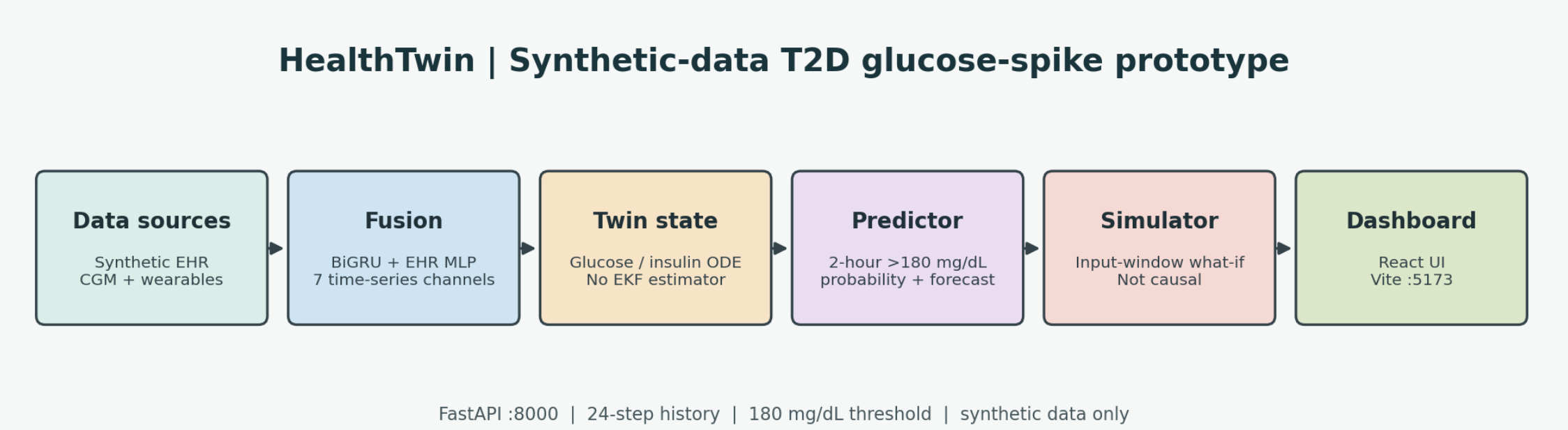
# Healthcare use case | India

- Clinician-supervised review: patient history → risk estimate → factor review → what-if exploration.
- Indian meal examples and Asian-Indian BMI cutoffs are intended context features.
- No clinician workflow integration or clinical evaluation has been performed.

# Data and cohort

- Procedurally generated synthetic EHR and 5-minute CGM/wearable traces.
- Generators describe a 500-patient cohort; the checked-in demo bundle uses P0001 only.
- Wearable relationships and physiology are simplified assumptions, not validated data.

# System architecture



# Physiological twin

- Bergman-inspired glucose, insulin, remote-action, and gut compartments.
- Inputs include carbohydrates, activity, sleep, stress, and medication adherence.
- The current code does not include the EKF/Bayesian estimator mentioned in planning notes.

# Predictor and features

- 24 history steps × 7 channels plus 12 static EHR features.
- Bidirectional GRU and EHR MLP feed a fusion head for classification and four forecast horizons.
- XGBoost and logistic regression are tabular baselines.

# Stored results snapshot

| | AUROC | AUPRC | F1 | ECE |
|---|---|---|---|---|
| **GRU (glucose only)** | 0.8142 | 0.8458 | 0.6876 | 0.0540 |
| **GRU + Wearables (no EHR)** | 0.9080 | 0.9128 | 0.8136 | 0.0455 |
| **Full Fusion (GRU + EHR)** | 0.9157 | 0.9221 | 0.8186 | 0.0582 |
| **XGBoost (tabular)** | 0.9417 | 0.9452 | 0.8501 | 0.0376 |
| **Logistic Regression** | 0.9059 | 0.9092 | 0.8108 | 0.0340 |

Values are read from results/metrics.json; they are not reproduced by current all-negative labels.

# Metrics validity and uncertainty

- The metrics artifact reports 239,316 train windows and 58.31% positives.
- Current label arrays have 997,150 / 213,675 / 213,675 entries and contain only zeros.
- Metrics are not reproducible from current arrays; no confidence intervals are stored.

# Simulator demonstration

- Select the synthetic patient P0001.
- Inspect history and factors, then submit a carbohydrate/activity/medication what-if.
- Displayed changes are model simulations, not causal effects or treatment advice.

# Robustness status

- Activity and carbohydrate monotonicity checks are documented as failing.
- No 5-seed confidence intervals, distribution-shift suite, or external validation result is present.
- Release gate: rebuild data, models, and metrics together, then rerun the checks.

# Explainability and guardrails

- Factor groups are normalized for display; importance is not causal attribution.
- Risk bands are model-defined, not medical standards.
- The UI and API carry synthetic-data and non-clinical disclaimers.

# Limitations

- Synthetic-only cohort and rule-based wearables.
- Stored metric/data mismatch; forecast performance and clinical utility are unverified.
- No real-world safety, privacy certification, or prospective validation.

# Potential impact

- A future validated tool could help teams review risk trajectories and intervention hypotheses.
- This prototype supports technical exploration only.
- Any clinical study requires governance, consent, and prospective evaluation.

# Roadmap and release gates

- Regenerate the synthetic cohort and check label, split, and timestamp integrity.
- Retrain all models; reproduce metrics with patient-level confidence intervals and stress tests.
- Review the demo with clinicians, finalize public artifacts and team details, then publish.